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LIME vs SHAP: A Paired Empirical Comparison

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Abstract

LIME and SHAP are widely used explainability methods, yet guidance on when each is preferable remains limited and often anecdotal. This paper presents a focused paired empirical comparison of the two methods on tabular data using five primary technical metrics under a controlled benchmarking protocol. Across 75 matched SHAP-LIME cells, SHAP dominates on stability, fidelity, and faithfulness gap, while LIME yields systematically sparser explanations and lower runtime in most contexts. The cost trade-off is statistically significant but heterogeneous across model families: matched-cell medians are 684.1 ms for SHAP versus 65.7 ms for LIME, whereas means are 11,708.3 ms versus 3,660.7 ms because kernel-SHAP tails are heavy. These findings support objective-driven deployment guidance: SHAP for high-assurance auditing and LIME for latency-sensitive user-facing explanation workflows.

Keywords: explainable ai, lime, shap, interpretability, robustness benchmarking

1 Introduction

Selecting an explanation method in production can be posed as a constrained decision problem: given a fixed predictor f and instance x, choose $E \in \{\text{LIME}, \text{SHAP}\}$ to maximize explanation quality while satisfying operational constraints. Concretely, we consider

$$E^\star = \arg \max_{E \in \{\text{LIME,SHAP}\}} U(S_E, F_E, \Delta_{k,E}, P_E) \quad \text{s.t.} \quad C_E \leq \tau,$$

where S is stability, F fidelity, Δ_k faithfulness gap, P sparsity/parsimony, C cost, and τ a latency budget.

LIME and SHAP optimize different principles. LIME explains locally by fitting a sparse surrogate around x,

$$\xi(x) = \arg \min_{g \in G} \mathcal{L}(f, g, \pi_x) + \Omega(g),$$

favoring local fidelity and explicit sparsity controls (Ribeiro et al., 2016). SHAP explains predictions through additive feature attributions $f(x) = \phi_0 + \sum_i \phi_i$, where ϕ_i are Shapley-based contributions with game-theoretic guarantees (Lundberg and Lee, 2017). These theoretical differences are well known; however, they do not by themselves resolve deployment choices across model families, sampling regimes, and runtime constraints (Retzlaff et al., 2024; Ali et al., 2023).

Recent synthesis and systematic studies show that XAI evaluation remains fragmented, with no single accepted protocol and strong dependence on which evaluation dimensions are prioritized (Longo et al., 2024; Ali et al., 2023; Pawlicki et al., 2024; Canha et al., 2025). This makes single-metric comparisons insufficient for claim-ready method selection.

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Open benchmarking toolchains and robustness-aware faithfulness protocols further show that method rankings can change under different metric families and perturbation settings (Agarwal et al., 2022; Sithakoul et al., 2024; Zheng et al., 2025).

Within the broader benchmark program, this manuscript isolates the matched LIME-versus-SHAP decision problem rather than repeating the full multi-method framework contribution. This narrower scope allows a more detailed paired analysis of quality-cost trade-offs across model families and runtime regimes.

This paper provides a paired, multi-objective benchmark using an audited, configuration-driven protocol on tabular data (UCI Adult). We evaluate five technical endpoints with explicit directionality: Stability (S, higher is better), Fidelity (F, higher is better), Faithfulness Gap (Δ_k, higher is better under top-k masking), Sparsity (active-feature ratio P, lower is better for parsimony), and Computational Cost (C, lower is better). Inference is performed on matched SHAP-LIME cells with identical $(\text{model}, \text{seed}, N)$, using paired Wilcoxon tests (Wilcoxon, 1945), paired effect sizes (d_z) (Lakens, 2013), and Holm correction (Holm, 1979).

Our paired analysis indicates a consistent trade-off: SHAP tends to provide stronger stability, fidelity, and top-k faithfulness signals, while LIME offers systematically sparser explanations and substantially lower runtime cost in most settings. This supports objective-driven method selection rather than universal method supremacy (Agrawal et al., 2025).

Our contributions are:

1. **Formalized deployment trade-off**: we frame LIME-vs-SHAP as a constrained multi-objective selection problem over quality, parsimony, and cost.
2. **Paired inferential evidence**: we quantify method differences on matched cells with corrected significance testing and effect sizes for each endpoint.
3. **Operational heterogeneity analysis**: we characterize when cost dominance changes by model family, including heavy-tail latency behavior.
4. **Decision guidance**: we derive deployment recommendations that map system constraints to explainer choice.

2 Background

This section summarizes the two explainability methods compared in our benchmark and highlights the technical properties that directly motivate the evaluation metrics and hypotheses used in this paper.

2.1 LIME (Local Interpretable Model-Agnostic Explanations)

LIME approximates a black-box model f locally around an instance x by training an interpretable surrogate model g (e.g., linear regression) on perturbed samples weighted by a kernel π_x. It solves:

$$\xi(x) = \arg\min_{g \in G} \mathcal{L}(f, g, \pi_x) + \Omega(g).$$

where $\Omega(g)$ penalizes complexity (sparsity) (Ribeiro et al., 2016). In practice, LIME behavior is strongly shaped by its neighborhood construction and surrogate-design hyperparameters

(e.g., kernel width, number of perturbation samples, and feature budget). These controls often yield compact explanations and low latency, but they also introduce stochastic variation because local neighborhoods are sampled rather than exhaustively enumerated.

2.2 SHAP (Shapley Additive Explanations)

SHAP attributes prediction output to features based on their marginal contribution across all possible coalitions, satisfying axioms of efficiency, symmetry, and dummy properties (Lundberg and Lee, 2017). KernelSHAP estimates these quantities with a weighted regression scheme over coalitions for model-agnostic usage, whereas tree-specific implementations can exploit model structure. Relative to local-surrogate methods, SHAP is typically denser and computationally heavier in non-tree settings because many coalition evaluations are required to estimate stable attributions.

2.3 Implications for This Benchmark

For a fair LIME-versus-SHAP comparison, the benchmark must control confounders unrelated to explainer design. This motivates the protocol choices implemented in Section 3: identical transformed feature space, shared model artifacts, fixed class target, and matched paired cells by $(\text{model}, \text{seed}, N)$. Under these controls, expected methodological trade-offs become testable: LIME should favor sparsity and cost, while SHAP should favor stability and faithfulness-oriented quality. This framing also aligns with recent calls for multi-metric, functionally grounded XAI evaluation rather than single-endpoint reporting (Pawlicki et al., 2024; Canha et al., 2025).

3 Methodology

This study uses a configuration-driven experimental pipeline for reproducible XAI benchmarking. The pipeline fixes (i) preprocessing and model artifacts before explanation, (ii) explainer hyperparameters via versioned YAML configurations with deterministic seeds, (iii) artifact-level logging per run (including `results.json`), and (iv) explicit validity filters plus matched SHAP-LIME pairing for inferential analysis. This structure follows recent recommendations for functionally grounded, multi-metric XAI evaluation and transparent benchmarking (Canha et al., 2025; Pawlicki et al., 2024; Agarwal et al., 2022). We instantiate this pipeline for a focused LIME-versus-SHAP comparison to produce claim-ready evidence on quality-cost trade-offs under controlled conditions.

3.1 Research Questions and Hypotheses

The methodology is model-agnostic and reproducibility-oriented, with all confirmatory claims anchored in quantitative endpoints.

We address the following questions:

1. **RQ1 (Robustness)**: Is SHAP more stable than LIME under input perturbation?
2. **RQ2 (Parsimony)**: Is LIME sparser than SHAP?
3. **RQ3 (Efficiency)**: Is LIME computationally cheaper than SHAP?

4. **RQ4 (Faithfulness)**: Does SHAP provide stronger faithfulness/fidelity signals than LIME?

Corresponding directional hypotheses are:

1. $H_1 : S_{\text{SHAP}} > S_{\text{LIME}}$ (stability).
2. $H_2 : P_{\text{LIME}} > P_{\text{SHAP}}$ (sparsity).
3. $H_3 : C_{\text{LIME}} < C_{\text{SHAP}}$ (cost).
4. $H_4 : F_{\text{SHAP}} > F_{\text{LIME}}$ (fidelity/faithfulness).

3.2 Experimental Design and Analysis Units

The robustness cohort is instantiated from the repository-tracked EXP2-scaled matrix with factors:

- model family $g \in \{\text{logreg}, \text{rf}, \text{xgb}, \text{svm}, \text{mlp}\}$,
- explainer $k \in \{\text{lime}, \text{shap}\}$,
- seed $s \in \{42, 123, 456, 789, 999\}$,
- sampling intensity $N \in \{50, 100, 200\}$ per TP/TN/FP/FN stratum.

This defines $5 \times 2 \times 5 \times 3 = 150$ planned LIME/SHAP runs. In the current artifact state, we retain 150 analyzable LIME/SHAP runs (75 LIME, 75 SHAP) and 75 matched SHAP-LIME cells on identical (g, s, N) coordinates for paired inference. Accordingly, this manuscript reports the EXP2 robustness cohort rather than the earlier EXP1 calibration subset, because paired multi-model coverage is required for inferential claims.

To avoid pseudoreplication, we use a hierarchical analysis structure:

1. **Instance level**: per-case metric values.
2. **Run level**: mean metric per configuration.
3. **Matched-cell level**: paired SHAP-LIME run means on shared (g, s, N) cells (primary inferential unit).

3.3 Data, Preprocessing, and Model Controls

All experiments use the UCI Adult data set (Kohavi, 1996) with a stratified split and deterministic seeds. The preprocessing pipeline follows train-only fitting with:

- numeric scaling (‘StandardScaler‘),
- categorical one-hot encoding (‘OneHotEncoder‘ with `handle_unknown='ignore'`).

Pre-trained black-box models are loaded from fixed artifacts and reused across explainers to isolate explainer effects: random forest (Breiman, 2001) and gradient-boosted trees (`xgb`) (Chen and Guestrin, 2016), alongside additional tabular families present in the robustness cohort.

Evaluation instances are sampled by error quadrants (TP, TN, FP, FN) via stratified sampling. Nominal run size is $4N$, with realized counts bounded by quadrant availability and runtime failures (observed range: 27 to 800 instances; median 400).

3.4 Explainer Protocol

Both explainers consume the same transformed feature space and target the positive class probability:

- **SHAP**: TreeExplainer for tree models, KernelExplainer for non-tree models, with background subsampling ($n = 50$).
- **LIME**: tabular local surrogate with `num_features=10`, effective `num_samples=1000`, and `kernel_width=3.0` in the EXP2 configuration family.

3.5 Metric Operationalization

Primary metrics are computed per instance and aggregated per run:

1. **Stability** S: mean pairwise cosine similarity of attribution vectors under Gaussian perturbations ($T = 15$, $\sigma = 0.1$):

$$S = \frac{2}{T(T-1)} \sum_{a<b} \cos\Big(e^{(a)}, e^{(b)}\Big) .$$

2. **Sparsity** P: active-feature ratio above threshold $\tau = 10^{-4}$ (lower active ratio indicates sparser explanations).
3. **Fidelity** F: faithfulness correlation between attribution magnitude and prediction drop under feature masking.
4. **Faithfulness Gap** Δ_k: absolute prediction change after masking top-k features ($k = 5$):

$$\Delta_k = \big| f(x) - f(x_{\text{mask-top-k}}) \big| .$$

5. **Cost** C: per-instance wall-clock explanation time (ms).

3.6 Inferential Protocol

The statistical plan follows established comparative-ML practice (Demsšar, 2006) and thesis ADR decisions:

1. primary test: two-sided paired Wilcoxon signed-rank on matched SHAP-LIME cells for each metric (Wilcoxon, 1945),
2. effect size: paired d_z for magnitude reporting (Lakens, 2013),

Table 1: Matched SHAP-versus-LIME paired results ($n = 75$ cells). For all metrics, $\Delta =$ SHAP - LIME. Sparsity is measured as active-feature ratio, so larger values indicate denser explanations.

Metric	LIME Mean	SHAP Mean	Δ	p_{Holm}	d_z
Stability	0.014	0.732	+0.718	2.6×10^{-13}	3.00
Sparsity (active ratio)	0.085	0.226	+0.142	2.6×10^{-13}	0.84
Fidelity	0.560	0.808	+0.248	2.6×10^{-13}	4.82
Faithfulness Gap	0.334	0.380	+0.045	2.6×10^{-13}	2.63
Cost (ms)	3660.7	11708.3	+8047.6	1.2×10^{-10}	0.45

3. multiplicity control: Holm-Bonferroni over the five primary metrics (Holm, 1979).

All inferential claims are restricted to artifact-qualified runs: parseable, non-empty results with explicit matched pairing. No semantic endpoint enters the confirmatory analysis reported here.

3.7 Reproducibility Contract

The protocol is fully configuration-driven (YAML + seeded execution), with per-run ‘results.json‘ artifacts, explicit exclusion of malformed or empty runs, and no synthetic imputation. This provides a deterministic audit trail from configuration to manuscript-level claim.

4 Results

Results are reported from the artifact-qualified EXP2 robustness cohort. In the current merged recovery snapshot, the EXP2 analysis exports contain 252 analyzable unique runs out of 300 planned cells overall. For this paper's paired LIME-versus-SHAP inference, we use the fully matched subset: 150 analyzable runs (75 LIME, 75 SHAP) forming 75 paired cells with identical $(\text{model}, \text{seed}, N)$. Matched coverage spans all five model families (xgb = 15, logreg = 15, rf = 15, mlp = 15, svm = 15) and all sampling intensities $N \in \{50, 100, 200\}$ with near-balanced counts.

4.1 Primary Paired Inference

Table 1 summarizes paired run-level outcomes on the 75 matched cells. All five primary endpoints remain significant under Holm-adjusted testing. All numeric entries in Table 1 are source-locked to the analysis export `wilcoxon_shap_lime_all_models.csv` under `outputs/analysis/paper_a_exp2_stats/`.

The stability effect is the largest practical separation in the study. Median paired stability difference is +0.866 in favor of SHAP, and the direction is consistent in all 75 matched cells. This indicates that LIME's local surrogate sampling introduces substantial run-to-run volatility under perturbation, whereas SHAP is comparatively invariant.

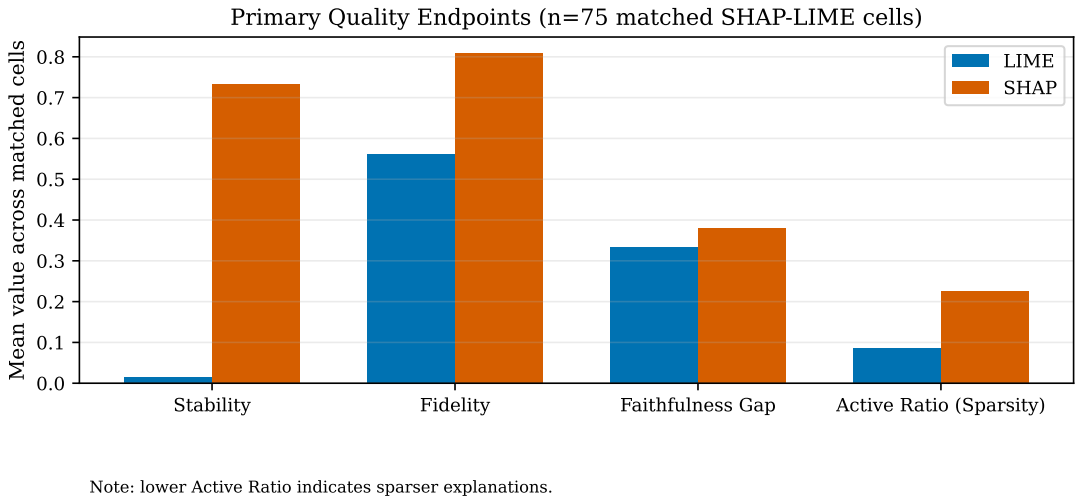

Figure 1: Paired mean comparison across quality-oriented primary endpoints ($n = 75$ matched SHAP-LIME cells). Active ratio is reported directly; lower values imply sparser explanations.

For sparsity, the sign is also consistent in all 75 matched cells, but interpretation requires care: the metric here is active-feature ratio. SHAP's larger value means denser explanations; thus LIME is systematically sparser and therefore easier to compress for user-facing narratives.

Fidelity favors SHAP in all 75 matched cells, with a mean paired difference of +0.248 and a very large paired effect size ($d_z = 4.82$). Faithfulness gap also favors SHAP in 74 of 75 matched cells, with a mean paired difference of +0.045 and a large paired effect size ($d_z = 2.63$). These results indicate not only statistical separation but practically meaningful quality gains under this protocol. Figure 1 visualizes these paired mean separations for the four quality-oriented endpoints.

4.2 Runtime Distribution and Heterogeneity

The cost signal is significant but strongly heterogeneous. SHAP is slower in 59 of 75 matched cells. Matched-cell median cost is 684.1 ms for SHAP versus 65.7 ms for LIME, corresponding to a median SHAP/LIME cost ratio of approximately 5.3×. Means are more separated (11,708.3 ms versus 3,660.7 ms) because the SHAP distribution is heavy-tailed: the 90th and 95th percentile latencies are approximately 51,456 ms and 67,066 ms, respectively, compared with 12,825 ms and 13,192 ms for LIME. This tail behavior is concentrated in kernel-SHAP contexts, especially SVM, which inflates mean cost beyond median behavior. Figure 2 shows both the overall log-scale runtime distribution and model-level median heterogeneity.

Heterogeneity analysis by model family shows that quality ordering (stability, sparsity-direction, fidelity, faithfulness gap) is robust, while cost ranking is model dependent. In particular, SHAP is faster in all 15 XGBoost matched cells and in one SVM cell, whereas

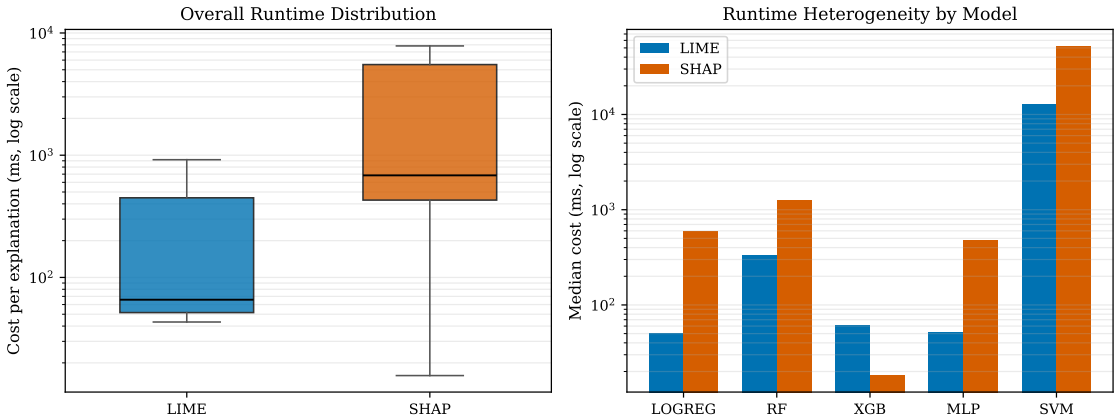

Figure 2: Runtime behavior under matched-cell pairing ($n = 75$). Left: overall cost distribution by method (log scale). Right: model-level median cost (log scale), showing the XGBoost exception where SHAP is faster while other families are SHAP-slower.

LogReg, RF, and MLP are uniformly SHAP-slower. Therefore, "LIME is faster" is a majority statement in this cohort, but it is not universally true across model classes.

4.3 Synthesis of Trade-offs

The empirical frontier is clear: SHAP delivers higher reliability and faithfulness at substantially higher expected runtime, while LIME delivers lower latency and stronger compression at the cost of robustness. The results therefore support objective-driven selection rather than global method supremacy. For assurance-critical analysis, SHAP's quality margin is large and consistent. For interaction-critical systems, LIME's runtime profile remains operationally attractive.

5 Practical Recommendations

Based on these empirical findings, we propose the **Hybrid Deployment Pattern**:

1. **Tier 1: Real-Time / End-User (Use LIME)**
   - Use when explaining predictions to end-users in a live application (e.g., "Why was my loan denied?" in a mobile app).
   - Evidence basis: lower matched median runtime (65.7 ms) and systematically lower active-feature ratio under the current protocol.
   - Recommended use condition: interactive settings where sub-second latency and compact explanations matter more than maximizing fidelity.
   - Caveat: XGBoost is an exception in this cohort, where SHAP is consistently faster than LIME.
2. **Tier 2: Auditing / Data Science (Use SHAP)**

- Use when analyzing model bias, debugging model logic, or responding to regulatory inquiries.
- Evidence basis: consistent paired advantages in stability, fidelity, and faithfulness gap across the matched cohort.
- Recommended use condition: analyst-mediated or offline workflows where second-level or minute-level latency is acceptable.
- Caveat: runtime costs depend strongly on whether SHAP operates in tree-specific or kernel mode.

6 Validity Boundaries

6.1 Statistical and Construct Validity

This manuscript's confirmatory claims are restricted to the 75 fully matched SHAP-LIME cells in the EXP2 robustness cohort. Unlike the full multi-method benchmark reported in Paper A, the pairwise subset used here is claim-complete after artifact qualification: 75 LIME runs and 75 SHAP runs remain available for paired inference. Reported metrics are implementation-bound rather than abstract labels; their operational definitions are anchored in `src/metrics/fidelity.py`, `src/metrics/stability.py`, `src/metrics/sparsity.py`, `src/metrics/faithfulness.py`, and `src/metrics/cost.py`. No semantic, human-utility, or LLM-judge claim enters the confirmatory scope of this paper.

6.2 Internal and External Validity

Runtime comparisons are protocol-specific. SHAP combines TreeExplainer for tree models and KernelExplainer for non-tree models, whereas LIME is run with fixed local-surrogate settings (`num_features=10`, effective `num_samples=1000`, `kernel_width=3.0`). Accordingly, the reported cost gap reflects both method family and concrete implementation pathway. External generalization beyond Adult tabular classification is not established; rankings may shift under different datasets, modalities, feature transformations, masking schemes, or hyperparameter settings.

6.3 Reproducibility and Operational Validity

All reported claims are traceable to seeded YAML configurations, per-run `results.json` artifacts, explicit exclusion of malformed or empty runs, and inferential exports under `outputs/analysis/paper\_a\_exp2\_stats/`. The current merged recovery snapshot contains 252 analyzable unique runs overall, but the Paper B subset is fully matched and therefore sufficient for the SHAP-versus-LIME claims made here. Figures are regenerated directly from `wilcoxon\_shap\_lime\_all\_models.csv` and `paired\_cells\_shap\_lime\_all\_models.csv`, and no synthetic imputation is used at any stage.

7 Limitations and Future Work

- **Data Set Scope**: This study focuses on tabular data (Adult). Image and text domains may exhibit different trade-offs.

- **Configuration Space**: We compared standard configurations (KernelSHAP versus standard LIME). Hyperparameter tuning (e.g., LIME kernel width) can alter results.
- **Runtime Heterogeneity**: Cost differences are strongly model-family dependent; the XGBoost exception shows that "LIME is faster" should not be interpreted as a universal rule.
- **Human-Centered Utility**: This paper does not establish human-grounded usefulness or semantic clarity; validated user studies or human-calibrated semantic evaluation remain future work.
- **Next Extensions**: Replication on additional datasets and integration of thesis-planned causal or counterfactual endpoints remain open follow-up tasks rather than current contributions.

8 Code and Artifact Availability

The current manuscript draft and supporting materials are available in the public repository at `https://github.com/jonnabio/xai-eval-framework`. Materials needed to reproduce Paper B include:

- `experiments/exp2\_scaled/results/`
- inferential export directory: `outputs/analysis/paper\_a\_exp2\_stats/`
- key export: `analysis_summary.json`
- key export: `wilcoxon_shap_lime_all_models.csv`
- key export: `paired_cells_shap_lime_all_models.csv`
- `scripts/run\_exp2\_statistical\_analysis.py`
- `scripts/generate\_paper\_b\_figures.py`
- manuscript sources under `docs/reports/paper\_b/`

The repository is released under the MIT License. Before journal submission, the exact review snapshot should be frozen as a versioned release and archived under a version-specific DOI.

9 Conclusion

The choice between LIME and SHAP is not a matter of "better," but of "fit for purpose." Under this protocol, SHAP is the preferred choice when stability and fidelity-oriented quality dominate the objective, while LIME remains the pragmatic option when latency and compactness are the primary constraints. Our paired analysis across 75 matched robustness cells shows consistent SHAP advantages in stability, fidelity, and faithfulness gap, alongside systematic LIME advantages in sparsity and lower runtime for most (but not all) model

contexts. This evidence supports architecture-level deployment patterns that separate user-facing low-latency explanation paths from high-assurance auditing workflows.

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